

Ritumoni Sarma

Senior Staff Frontend Engineer · Platform Engineering & AI Systems

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PROFESSIONAL SUMMARY

Senior Staff Frontend Engineer and hands-on platform builder with 13+ years of experience delivering high-scale UI systems, Micro-Frontend (MFE) architectures, and AI-augmented developer tooling including RAG Pipelines and Agentic AI workflows. Deep technical owner across cross-functional platform initiatives — generating \$1.6M in infrastructure savings, recovering 2,500+ engineering hours annually, and driving 79% performance improvements and 91% latency reductions for global FinTech and SaaS platforms. Core expertise in React platform engineering, RFC/ADR-driven technical decision making, and frontend performance engineering at scale.

TECHNICAL SKILLS

Architecture:	Micro-Frontends (MFE), Module Federation, UI System Design, Distributed Systems, Monorepo, Nx, Turborepo, Domain-Driven UI, Atomic Design
Engineering:	React (Internals, Reconciler, Concurrent Mode), React Query, TanStack, React Router, Zustand, Context API, Custom Hooks, useMemo, React Profiler, TypeScript, ES6+, Node.js, Next.js, Redux, GraphQL, REST APIs
AI & Emerging:	Agentic AI, RAG Pipelines, RAG-integrated Frontends, Prompt Engineering, LLM Integration, LangChain, Structured LLM-to-UI Chaining
Infra & Tooling:	Webpack, Vite, AWS (S3, CloudFront, Lambda@Edge), Docker, CI/CD (GitHub Actions), Jest, Playwright, Storybook, Figma, Observability
Performance:	Core Web Vitals, Code Splitting, Tree Shaking, Lazy Hydration, Performance Budgets, Rollback Automation, CDN Orchestration

LEADERSHIP & PROCESS

Technical Lead:	RFC/ADR Authoring, Platform Engineering, Cross-functional Collaboration, Technical Mentorship, Design System Governance, Staff Engineering
Culture:	Formal Mentorship (30+ Engineers), TDD Institutionalization, Code Review Standards, Agile, Scrum
Compliance:	WCAG / Accessibility, CSP, Dependency Auditing, Supply-Chain Integrity

PROFESSIONAL EXPERIENCE

AlgoDuel Technical Services Senior Staff Frontend Engineer Bengaluru, India	Oct 2025 – Present
- Designed and built a RAG-based AI Discovery Engine indexing 5,000+ code artifacts, reducing developer onboarding and discovery time by 60% through close collaboration with cross-functional stakeholders. - Built Agentic AI pipelines for automated code-health triage, pattern detection, and technical debt prioritization, recovering 2,500+ senior engineering hours annually. - Authored the ADR/RFC decision framework for AI integration, adopted by 10+ squads to ensure architectural consistency and zero cross-team integration failures during rapid platform scaling. - Engineered production-grade prompt pipelines using structured LLM-to-UI output chaining, reducing initial UI interaction latency by 40% for global developer tooling traffic.	
Celigo Senior Staff Frontend Engineer Bengaluru, India	Aug 2023 – Sep 2025
- Owned and executed the migration to a Federated MFE Architecture, decoupling 11 business units and enabling independent, zero-downtime release cycles across a global SaaS portfolio. - Delivered 79% improvement in Core Web Vitals (11s to 2.3s) through advanced route-based code splitting and lazy hydration, directly improving customer retention metrics. - Drove technical debt prioritization across the platform, producing a 35% reduction in recurring bug reports and a 58% global bundle size reduction. - Designed and implemented org-wide security hardening including strict CSP policies and supply-chain integrity checks across 11 business units.	
Upstox Staff Frontend Engineer — Trading Systems Mumbai, India	Feb 2023 – Aug 2023

- Drove 21% growth in user engagement by reducing interaction latency 91% (450ms to 40ms) on a high-frequency trading platform.
- Designed and built a real-time visualization engine rendering 30,000+ live data points at 60 FPS with zero frame drops for derivatives trading.
- Applied React Profiler, useMemo, and virtualization patterns to systematically eliminate render bottlenecks across a mission-critical trading UI.

Lowe's Companies, Inc. | Tech Lead, Frontend Engineering

Aug 2021 – Feb 2023

Bengaluru, India

- Generated \$1.6M in annual infrastructure savings through a 68% reduction in critical-path load times and optimized cloud delivery patterns.
- Defined and maintained frontend performance standards and architectural guidelines adopted across 17+ engineering teams.
- Built and ran a formal mentorship program for 30+ engineers, institutionalizing TDD, architecture walkthroughs, and RFC-driven technical decision making.

Fidelity International | Senior Software Engineer

Sep 2018 – Aug 2021

Bengaluru, India

- Delivered 30% improvement in search relevance by designing and integrating an Elasticsearch-driven UI layer for global financial advisor workflows across 5+ international markets.
- Built modular, internationalized UI systems with high availability and full WCAG compliance.

EARLY CAREER HIGHLIGHTS

ARI Network Services | Senior Software Developer: Re-platformed legacy systems with Node.js and built a standardized UI component library adopted org-wide.

Goibibo | Frontend Engineer: Optimized high-traffic booking flows under constrained network conditions, increasing mobile conversion rates.

Tesco HSC | Software Developer: Delivered accessible, performant commerce interfaces for global retail operations.

EDUCATION

Bachelor of Engineering — Electronics & Telecommunications

RV College of Engineering, Bengaluru | 2012